Topic 7: Intermolecular Forces

7.1 — Types of Intermolecular Forces

Type of Force	Also Known As	Occurs Between
London Forces	Instantaneous dipole–induced dipole	All molecules, polar or non-polar
Permanent Dipole–Dipole Interactions	—	Polar molecules only
Hydrogen Bonding	Strong dipole–dipole with H bonded to N/O/F	Molecules with H–N, H–O, or H–F

1. London Forces (Dispersion Forces)

- Caused by **temporary fluctuations** in electron density → creates an **instantaneous dipole**
- Induces a dipole in a nearby molecule → **weak attraction**
- Present in **all** molecules
- **Strength increases** with:
  - **Number of electrons** (molecular size/mass)
  - **Surface area** (straight-chain alkanes > branched alkanes)

Example:
CH_4, Cl_2, I_2, noble gases (Ar, Xe)

2. Permanent Dipole–Dipole Interactions

- Occur between molecules with a **permanent dipole** (uneven charge distribution)
- Only exists in **polar molecules**
- Stronger than London forces but weaker than hydrogen bonding

Example:
HCl, CH_3Cl, SO_2

3. Hydrogen Bonding

- A strong form of dipole–dipole interaction
- Occurs when H is **covalently bonded to N, O, or F**, and this H is attracted to a lone pair on a nearby N, O, or F atom

- Strongest intermolecular force
- Causes **higher boiling points**, **solubility in water**, and **anomalous density**

Requirements:

1. Highly electronegative atom (N, O, F)
2. Lone pair on that atom
3. Hydrogen atom covalently bonded to it

Example:
H_2O, NH_3, HF

7.2 — Hydrogen Bonding in Specific Molecules (H_2O, NH_3, HF)

These molecules show **extensive hydrogen bonding** due to:

1. **Highly electronegative atoms**: N, O, or F
2. **Lone pairs** on these atoms
3. **Hydrogen** covalently bonded to N, O, or F

A. Water (H_2O)

- Oxygen is **highly electronegative**
- Each O atom has **2 lone pairs**
- Each H_2O can form **up to 4 hydrogen bonds** (2 donors, 2 acceptors)
- Strong hydrogen bonding network

Structure in ice:

- Forms a **tetrahedral open lattice**
- Causes ice to be **less dense** than water

Consequence:

- H_2O has a **much higher boiling point** than expected for its molar mass (18 g mol^{-1})

B. Ammonia (NH_3)

- Nitrogen has 1 lone pair and forms 3 N–H bonds
- Each NH_3 molecule forms **fewer hydrogen bonds** than H_2O (3 donors, 1 acceptor)
- Hydrogen bonding exists but is **less extensive**

Consequence:

- Higher boiling point than PH_3 but **lower than H_2O** (due to fewer H-bonds)

C. Hydrogen Fluoride (HF)

- Fluorine is the **most electronegative** element
- Strong H–F bond polarity and lone pair on F
- HF forms **linear hydrogen-bonded chains**

Consequence:

- HF has a **high boiling point** for its molar mass
- Boiling point is **lower than H_2O** because HF molecules can form fewer H-bonds (1 per molecule)

7.3 — Anomalous Properties of Water from Hydrogen Bonding

A. High Boiling and Melting Points

- Due to strong **intermolecular hydrogen bonds**
- A lot of energy is required to separate water molecules → high enthalpy of vaporisation
- Boiling point of H_2O (100 °C) is far higher than predicted by trends in Group 16 hydrides (e.g., H_2S, H_2Se)

Trend (Group 16 Hydrides):

Molecule	Mr	Boiling Point (°C)
H_2O	18	100
H_2S	34	−60
H_2Se	81	−41
H_2Te	130	−2

→ The **deviation in H_2O** is due to **hydrogen bonding**

B. Density of Ice vs Water

- In **ice**, water molecules are held in a **fixed hydrogen-bonded lattice**
- This structure is **open and less dense**
- On melting, the lattice collapses slightly → molecules come **closer together**

Consequence:

- Ice **floats** on water (ice is less dense than liquid water)

- Crucial for aquatic life in cold climates

7.4 — Predicting the Presence of Hydrogen Bonding

A. Rule for Hydrogen Bonding

A molecule will form hydrogen bonds if:

1. **H is covalently bonded** to **N, O, or F**
2. The molecule has a **lone pair** on the N, O, or F atom to form an attraction with H

B. Examples of Molecules That Exhibit Hydrogen Bonding

Molecule	Explanation
CH_3OH (methanol)	O–H group present → H-bonding
$CH_3CH_2NH_2$ (ethylamine)	N–H bond present → H-bonding
HF	H–F bond present → strong H-bonding
Carboxylic acids (e.g., CH_3COOH)	O–H and C=O allow **dimer formation** via H-bonds
Amides (e.g., CH_3CONH_2)	N–H and lone pairs on O allow strong hydrogen bonding

C. Molecules That Do Not Form Hydrogen Bonds

Molecule	Reason
CH_3OCH_3 (dimethyl ether)	No O–H bond
CH_4, CCl_4, H_2S, PH_3	No N, O, or F bonded to H

Tip: Presence of electronegative atoms is not enough — you need an H–N, H–O, or H–F bond **and** a lone pair on the other molecule.

7.5 — Explaining Physical Properties Using Intermolecular Forces

A. Boiling Points of Alkanes (Trend with Chain Length)

- **London forces** increase with more electrons (greater molecular mass)
- Longer chains have **greater surface area** → stronger dispersion forces
- Boiling point increases

Example:
$CH_4 < C_2H_6 < C_3H_8 < C_4H_{10}$

B. Effect of Branching on Boiling Points (Alkanes)

- **Branched alkanes** have **lower boiling points** than straight-chain isomers

- Smaller surface area → weaker London forces

Example:
butane ($CH_3CH_2CH_2CH_3$) > methylpropane ($CH_3CH(CH_3)CH_3$)

C. Boiling Points: Alcohols vs Alkanes

Property	Alcohols	Alkanes
Intermolecular force	Hydrogen bonding	Only London forces
Boiling point	Higher	Lower
Volatility	Lower	Higher

Conclusion: Alcohols are **less volatile** due to strong hydrogen bonds.

D. Boiling Point Trend: Hydrogen Halides (HF to HI)

Factors involved:

1. **Polarity** (for permanent dipoles)
2. **Hydrogen bonding** (only HF)
3. **London forces** (increase with size)

Boiling point trend:
HF > HI > HBr > HCl

Molecule	Notes
HF	Hydrogen bonding (very high bp)
HI	Strong London forces (large Mr = 128)
HBr, HCl	Only dipole-dipole and London

7.6 — Understanding Solvent Choice Based on Intermolecular Forces

A. Types of Solvent–Solute Interactions

Type of Solvent	Main Intermolecular Forces Present	Example Solvents
Polar solvents	Hydrogen bonding and permanent dipoles	Water, ethanol
Non-polar solvents	London dispersion forces	Hexane, cyclohexane

B. Solubility of Ionic Compounds in Water

Reason:

- Water is a **polar solvent**.
- It forms **ion–dipole attractions** with cations and anions.
- These interactions **hydrate** the ions and overcome the lattice energy of the ionic compound.

Example:
$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$
Water molecules surround ions to stabilise them.

Exception:
Some ionic compounds (e.g., $BaSO_4$, AgCl) have very **high lattice enthalpies** and are **insoluble**.

C. Solubility of Alcohols in Water

- Alcohols (e.g., CH_3OH, CH_3CH_2OH) contain **O–H** groups → can form **hydrogen bonds** with water.
- Short-chain alcohols are **fully miscible**.
- As chain length increases, the non-polar part dominates → **decreased solubility**.

D. Water as a Poor Solvent for Polar Molecules (e.g., Halogenoalkanes)

- Halogenoalkanes are **polar**, but cannot form **hydrogen bonds** with water.
- Water cannot surround or stabilise them effectively.
- Intermolecular attractions are **weaker than those between water molecules**, so halogenoalkanes are **insoluble**.

Example:
CH_3CH_2Cl is insoluble in H_2O despite being polar.

E. Use of Non-Aqueous Solvents

Used when the solute has **intermolecular forces similar to the solvent**.

Solute Type	Solvent Used	Reason
Non-polar (e.g., I_2)	Cyclohexane, hexane	London forces match
Polar, no H-bonding	Propanone (acetone)	Dipole–dipole matches
Oils and fats	Organic solvents	Hydrophobic, non-polar

Example:

- Iodine dissolves well in cyclohexane (purple solution) but poorly in water.
- Organic reactions often use **ethanol** or **propanone** as solvents.

F. Summary: Factors Affecting Solubility

Solute–Solvent Compatibility	Outcome
Similar intermolecular forces	Solute dissolves
Different intermolecular forces	Solute is poorly soluble